CALL FOR COLLABORATION

Huaravito Reserve LTEMS

Long-Term Ecological Monitoring Site

Piedades Sur, Costa Rica | 10.1065°N, 84.5666°W | 1,250 m.a.s.l.

Project:

"The 10-Degree Memory: Characterizing Biogenic Thermal Decoupling in a Mature Premontane Forest Fragment"

Huaravito Reserve LTEMS (Long-Term Ecological Monitoring Site), located in Piedades Sur, Costa Rica (10.1065°N, 84.5666°W; 1,250 m.a.s.l.), is seeking academic and technical collaboration to investigate an exceptional biogenic cooling phenomenon.

1. The Preliminary Observation

Field measurements conducted in February 2026 documented an extreme thermal gradient of 10.2°C over approximately 150 meters horizontal distance. While an adjacent degraded clay extraction site reached 27.5°C under midday conditions, the mature forest interior ("The Shrine") maintained 17.3°C. If confirmed through systematic replication, this would represent one of the most substantial biogenic cooling effects documented in small tropical forest fragments (typical range: 3–8°C; Hardwick et al. 2015; de Frenne et al. 2019).

Key finding: A single forest fragment <0.6 hectares exhibits thermal decoupling comparable to extensive primary forest landscapes.

2. The Site: Mature Forest Structure with Old-Growth Characteristics

Available satellite imagery (1984–present via Restor/Global Forest Watch) and local oral testimony (50+ years) indicate that Huaravito's western sector has remained continuously forested. Structural attributes consistent with mature forest status include:

- Dominant overstory of old-growth Hauya elegans (Onagraceae) with extensive buttress roots
- >70% epiphytic colonization (bryophytes, orchids, ferns) on trunk surfaces
- Dense organic soil layer (20 cm depth) with active mycorrhizal networks
- 116 species documented via iNaturalist (2021–2026), including late-successional specialists

This "mature heart" appears to function as a high-efficiency biological cooling engine. Proposed mechanisms include canopy-mediated reduction of solar radiation, high epiphytic thermal inertia, evapotranspirative cooling, and deep organic layer insulation.

Research Question: Can small forest remnants with mature structure provide thermal microrefugia under regional warming scenarios?

3. Collaboration Opportunities

We seek partners (universities, research institutes, NGOs, individual scientists, graduate students) to contribute expertise and/or instrumentation in the following areas:

A. Microclimate Monitoring

- Installation of distributed temperature/humidity datalogger network (minimum 6 nodes: degraded baseline, forest edge, interior gradient)
- Anemometer array for wind velocity profiling (ground, mid-canopy, canopy top)
- Soil moisture probes at multiple depths (forest vs. degraded comparison)
- Infrared thermography of canopy and trunk surfaces

B. Structural & Functional Ecology

- Canopy cover quantification (hemispheric photography, LiDAR, or UAV-based photogrammetry)
- Leaf area index (LAI) estimation for Hauya elegans stands
- Epiphytic biomass quantification and species composition
- Sap flow measurements and evapotranspiration modeling
- Organic layer depth mapping and soil carbon analysis

C. Field Support & Capacity Building

- Undergraduate/graduate internships in tropical forest ecology
- Hands-on training in long-term ecological monitoring protocols
- Data management and iNaturalist/Terraware platform integration
- Collaborative manuscript preparation for peer-reviewed publication

4. What Huaravito Reserve LTEMS Offers Collaborators

✓ Access to a structurally mature forest fragment with documented thermal anomaly

The site provides a rare opportunity to study thermal regulation in a small, edge-exposed forest that nonetheless maintains primary forest-like microclimate conditions.

✓ 5-year baseline dataset

Continuous monitoring since 2021; 238 iNaturalist observations; 116 species documented; microclimatic data (2022–2025); validated extreme event response (February 2026 polar intrusion: +6°C buffering capacity).

✓ Multi-platform data infrastructure

iNaturalist (public validation), Terraware (forest inventory), Restor (site profile), FERM/FAO (global reporting), NASA GLOBE Observer (ground-truth for satellite validation).

✓ On-site collaboration with resident custodian

Daily site presence; intimate knowledge of seasonal patterns, species phenology, microhabitat variation; 5 years of continuous field observation.

✓ Biocultural research context

Sacred Natural Site designation; integration of traditional ecological knowledge; Earth Charter principles; community engagement potential.

✓ Logistical support

Site access, local ecological expertise, assistance with permitting (if required), potential lodging coordination.

5. Potential Research Outputs

Successful collaboration could yield:

- Peer-reviewed publication(s) in forest ecology, microclimate, or restoration science journals
- Contribution to SNAPP-MORE global dataset on restoration monitoring indicators
- Validation (or refinement) of thermal stability as a cost-effective proxy for ecosystem maturity
- Case study on climate refugia potential of small forest fragments in agricultural landscapes
- Undergraduate/graduate thesis projects

6. Timeline & Commitment

Immediate opportunities (2026):

- Thermal gradient replication study (dry season: Feb–Apr; wet season: Jun–Oct)
- Structural characterization (canopy cover, LAI, epiphyte surveys)

Medium-term (2026–2027):

- Continuous datalogger deployment (12-month minimum)
- Seasonal evapotranspiration measurements
- Comparative study with other premontane fragments in Costa Rica

Flexible collaboration models:

- Short-term field campaigns (1–4 weeks)
- Remote collaboration (data analysis, manuscript co-authorship)
- Long-term partnership (multi-year monitoring, graduate student placement)

7. Contact & Next Steps

Interested in exploring the interface between forest structure, microclimate regulation, and climate resilience?

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iNaturalist Project: inaturalist.org/projects/reserva-huaravito-sitio-natural-sagrado
Restor Profile: restor.eco/es/sites/c4c9b69f-1f0b-40bc-9d80-2d18fd1e54cf

Please include in initial contact:

- Your institutional affiliation and research background
- Specific expertise or instrumentation you could contribute
- Preferred collaboration timeline (short-term vs. long-term)
- Any questions about site access, logistics, or data availability

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"Decoding the thermal memory of mature forests to inform restoration and climate adaptation strategies."

February 2026